

llmXive Automated Review of Full Attention Strikes Back: Transferring Full Attention into Sparse within Hundred Training Steps

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a compelling method for sparse attention, but several factual claims regarding the experimental setup and reported numbers require verification to ensure they are supported by the evidence provided.

First, the experimental environment described in Section 5.1 lists “Python 3.14” and “PyTorch 2.8”. These versions do not currently exist in the public software ecosystem (as of late 2025/early 2026, the latest stable versions are Python 3.12/3.13 and PyTorch 2.5/2.6). This appears to be a hallucination or a significant typo. Since the reported speedups ($9.36\times$) are highly sensitive to the specific kernel implementations and library versions, this error casts doubt on the reproducibility of the results. The authors must correct these version numbers to match the actual environment used.

Second, the claim of “ $9.36\times$ prefill speedup at 1M context” and “ $2.01\times$ decode speedup” relies on a comparison against FlashAttention-2 (FA2). While the paper mentions using FA2 as a baseline, it does not explicitly detail the specific configuration of the baseline (e.g., whether it was the standard implementation or a specific optimized variant, the exact context length distribution, or the hardware utilization metrics). Given the magnitude of the speedup, it is critical to confirm that the baseline was run under identical conditions (e.g., same batch size, same precision, same kernel fusion settings) to ensure the comparison is fair and the numbers are accurate.

Third, in the “Results on Reasoning Tasks” section, the text states “inputs are extremely short (

0.93)”.*The syntax*”

0.93 ” is invalid LaTeX and likely a typo for a specific value (perhaps a ratio or a length in thousands). This ambiguity makes the claim about “extreme prompt-generation asymmetry” difficult to verify. The authors should correct this syntax and clearly state the metric being referenced.

Finally, Table 1 (LongBench) includes a row for “Full Attn” but the caption states “Full attention is excluded from this ranking.” The table underlines the second-best result among sparse methods, which is consistent with the caption, but the presence of the Full Attn row in the same table without a clear visual separation or note can be confusing. Ensuring the table formatting and caption are perfectly aligned will prevent reader confusion about whether the Full Attn result was part of the ranking logic.

These issues are primarily fixable by correcting the text and clarifying the experimental setup, but they are necessary to establish the factual accuracy of the paper’s claims.

Action items.

- **[writing]** The paper presents a compelling method for sparse attention, but several factual

claims regarding the experimental setup and reported numbers require verification to ensure they are supported by the evidence provided. First, the experimental environment described in Section 5.1 lists “Python 3.14” and “PyTorch 2.8”. These versions do not currently exist in the public software ecosystem (as of late 2025/early 2026, the latest stable versions are Python 3.12/3.13 and PyTorch 2.5/2.6). This appears

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure effectively visualizes performance gains but is critically flawed by a missing method name in the caption and legends, and the x-axis scales between the two panels are inconsistent, hindering direct comparison.

Figure 2

Figure 2 presents a clear heatmap of retrieval scores across layers and head indices, but lacks clarification on head indexing, model layer count alignment, and statistical grounding of the scores, which limits interpretability and scientific rigor.

Figure 3

The figure displays a standard training loss curve with clear axes ticks, but it lacks formal axis labels and a legend, relying on the caption for context.

Figure 4

The figure effectively displays accuracy and sparsity trends across sequence lengths with clear data labels. However, the caption is grammatically incomplete due to a missing subject, and the right y-axis label could be more precise regarding the definition of the sparsity metric.

Figure 5

The figure effectively communicates latency comparisons across different sequence lengths, but the caption is grammatically incomplete with a missing subject, and the y-axis categories are not explicitly defined in the legend.

Figure 6

Figure 6 provides a clear architectural overview but suffers from critical text omissions in the caption and diagram labels, and the Dynamic Top-p Selector lacks sufficient axis detail to interpret the selection mechanism.

Figure 7

The figure provides a clear and well-structured overview of the hardware-aware decoding kernel workflow. However, the caption contains a grammatical error with a missing model name, and the legend keys are not explicitly defined in the text.

Figure 8

The figure fails to substantiate its caption’s claim about semantic retrieval because it uses abstract placeholders instead of actual text or attention maps, and the accompanying bar chart lacks necessary labels and axes.

Figure 9

Figure 9 effectively illustrates diffuse retrieval but suffers from minor labeling inconsistencies and an incomplete legend that may confuse readers unfamiliar with the baseline.

Action items.

- **[fatal]** Figure 1: The caption and chart legends contain a placeholder ‘.’ instead of the method’s name (likely ‘RTPurbo’ based on the legend), rendering the figure’s subject undefined.
- **[science]** Figure 1: The right panel’s x-axis labels (128K, 192K, 256K, 360K, 512K) do not align with the left panel’s labels (32K, 64K, 128K, 256K, 512K, 1M), making direct comparison of efficiency and accuracy at specific context lengths difficult.
- **[writing]** Figure 1: The right panel’s y-axis label ‘Accuracy (%)’ is present, but the bars are not explicitly labeled as accuracy values in the legend or caption, relying on the reader to infer the metric from the axis.
- **[writing]** Figure 2: The y-axis label ‘Head Index’ is ambiguous; the caption mentions ‘query heads’, but the axis includes indices up to 31, which may exceed the number of query heads per layer in the model, or implies a flattened index without explanation.
- **[writing]** Figure 2: The x-axis label ‘Layer’ and tick labels (0–47) suggest 48 layers, but the model name ‘Qwen3-Coder-30B-A3B’ does not clarify if this matches the actual architecture; a note on layer count or model specs would improve clarity.
- **[science]** Figure 2: No statistical context (e.g., averaging method, number of samples, or variance) is provided for the ‘retrieval scores’; without this, the heatmap values lack interpretability regarding reliability or significance.
- **[writing]** Figure 3: The x-axis label ‘train/global_step’ is rendered as a grey text overlay on the plot area rather than a formal axis label, and the y-axis lacks a descriptive label (e.g.,

‘Loss’).

- **[writing]** Figure 3: The plot contains a single data series but lacks a legend to explicitly identify the curve, relying solely on the caption for context.
- **[writing]** Figure 4: The caption contains a missing subject (e.g., ‘Unlike baselines that collapse..., [METHOD NAME] sustains robust accuracy...’). The figure shows ‘RTPurbo’ in the legend, so the caption should explicitly name it.
- **[writing]** Figure 4: The right y-axis label ‘Sparsity (%)’ is ambiguous; the line values (94-98%) represent the percentage of tokens *kept* (or sparsity ratio), but ‘Sparsity’ usually implies the percentage of tokens *dropped*. Clarify if this is ‘Retention Rate’ or ‘Sparsity’.
- **[writing]** Figure 5: The caption contains a missing subject (e.g., ‘Sparse decoding speedup of [Model Name]’), making it grammatically incomplete and unclear.
- **[science]** Figure 5: The y-axis labels ‘Ours (Fused)’ and ‘Full Attn (FA2)’ are not explicitly defined in the legend; the legend only lists specific kernel components (e.g., ‘D16 Score + Top-p’), requiring the reader to infer the grouping.
- **[writing]** Figure 5: The x-axis label ‘Latency (μ s)’ is repeated for all three subplots, which is redundant and could be consolidated into a single label for the entire figure.
- **[fatal]** Figure 6: The caption reads ‘Overall architecture of .’ with a missing model name, and the diagram contains multiple instances of the same missing name (e.g., ‘sustains robust accuracy’ in Figure 4 caption context), indicating incomplete text rendering or copy-paste errors.
- **[science]** Figure 6: The ‘Dynamic Top-p Selector’ inset shows a cumulative curve with $p=0.9$ but lacks axis labels for the curve itself (e.g., what is plotted on y-axis vs x-axis beyond ‘Cumulative’ and ‘p’), making the selection rule ambiguous.
- **[writing]** Figure 7: The caption contains a missing model name (‘Overview of the hardware-aware decoding kernel in .’), likely due to a placeholder or formatting error.
- **[writing]** Figure 7: The legend items ‘Sort-free top-p’ and ‘Bandwidth-optimized’ are not explicitly defined in the caption, though their association with the red and green boxes is visually clear.
- **[science]** Figure 8: The figure displays abstract blocks labeled ‘A’, ‘B’, ‘C’ and a bar chart, but lacks the actual text passage or attention heatmap required to demonstrate ‘semantic relatedness’ or ‘similar patterns’ as claimed in the caption.
- **[science]** Figure 8: The orange arrows indicate a mapping from the first three blocks to the last three, but the middle section is obscured by ‘.....’, making it impossible to verify the ‘long-range’ or ‘far away’ context claim.
- **[writing]** Figure 8: The bar chart below the blocks has no axes, units, or legend to explain what the bar heights represent or the difference between orange and teal colors.

- **[writing]** Figure 9: The top panel’s x-axis label ‘Token position’ is not aligned with the tick marks, and the top-right y-axis label ‘Attn mass’ lacks a clear scale or unit definition in the caption.
- **[science]** Figure 9: The red curve in the bottom-left plot is labeled ‘Retrieval head’ but no corresponding legend entry exists for the gray ‘Uniform’ line, making comparison ambiguous without external context.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-written and avoids excessive in-group slang, but it relies on several CUDA-specific and systems-level terms that are undefined, creating barriers for a competent reader from an adjacent field (e.g., NLP, statistics, or general ML) who may not be familiar with GPU architecture internals.

Specifically, Section 4.1 introduces a dense string of CUDA terminology without glossing. Terms like “CTA” (Compute Thread Array), “SM” (Streaming Multiprocessor), and “half2” are used as if they are common knowledge. While standard for GPU kernel developers, they are not standard vocabulary for a general ML researcher. Similarly, the mathematical notation in the kernel description, specifically the “log-sum-exp pair (m_b, ℓ_b) ” and the variable p_{top} , lacks explicit definition in the text where they first appear. The reader is forced to infer the meaning of m_b and ℓ_b from context or prior knowledge of stable softmax implementations, which violates the self-containment requirement.

Additionally, the term “attention sinks” in Section 3.2 is used without a brief operational definition, relying solely on a citation to “streamingLLM”. While the concept is central to the method, a reader unfamiliar with that specific prior work would not know what a “sink” is in this context.

These are minor, text-only fixes. Adding one-sentence definitions or parenthetical glosses for these terms would make the paper fully accessible to the target audience without altering the scientific content.

Action items.

- **[writing]** Section 4.1 (Hardware-Aware Fast Top-p Decoding Kernel): The term ‘CTA’ (Compute Thread Array) is used without definition. While standard in CUDA programming, it is not universally known to all adjacent-field PhDs (e.g., those in NLP or statistics). Define it at first use: ‘CTA (Compute Thread Array, the basic execution unit in CUDA)’.
- **[writing]** Section 4.1: The symbol p_{exttop} appears in the text (‘cumulative attention mass reaches p_{exttop} ’) without prior definition. The parameter is referred to as ‘Top- p ’ or ‘ p ’ elsewhere (e.g., Table 1, Section 3.2). Define p_{exttop} explicitly as the cumulative probability

threshold (e.g., 0.9) where it first appears in Section 4.1.

- **[writing]** Section 4.1: The term ‘log-sum-exp pair (m_b, l_b) ’ is used without defining the variables m_b and l_b . An adjacent-field reader may not know these represent the maximum and log-sum-exp values used in the stable softmax computation. Add a brief clause: ‘...reduces it to a block-level log-sum-exp pair (m_b, l_b) , where m_b is the block maximum and l_b is the log-sum-exp value.’
- **[writing]** Section 4.1: The term ‘SM’ is used (‘allows the SM to maximize concurrent CTAs’) without definition. In CUDA context, this stands for Streaming Multiprocessor. Define it at first use: ‘...allows the SM (Streaming Multiprocessor) to maximize...’
- **[writing]** Section 4.1: The term ‘half2’ is used (‘vectorized *exttthalf2* instructions’) without explanation. While common in GPU optimization, it refers to a 2-element vector of 16-bit floats. A reader from a non-systems adjacent field might not know this. Add a brief gloss: ‘...via vectorized *exttthalf2* (2-element 16-bit float vector) instructions...’
- **[writing]** Section 3.2: The term ‘attention sinks’ is used in the phrase ‘sliding window with attention sinks’ without definition. While ‘streamingLLM’ is cited, the concept of ‘sinks’ (specific tokens that attract attention to stabilize the distribution) is not explained. Add a brief parenthetical: ‘...sliding window with attention sinks (specific tokens that attract attention to stabilize the distribution)...’

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s logical structure is generally sound: the conclusion that RTPurbo achieves efficient sparse inference follows from the premises of intrinsic head sparsity and low-dimensional retrieval. However, there are internal numerical inconsistencies in the experimental reporting that break the chain of reasoning regarding the training cost.

First, Section 5.1 states the second stage uses “about 600 steps” on a corpus with an “average length of 48K”. Appendix A.3.2 states the corpus contains “roughly 180M tokens”. Using the global batch size of 8 from Table A.2, the calculation $600 \times 8 \times 48,000$ yields 230.4M tokens, which directly contradicts the 180M figure. This discrepancy suggests either the step count, batch size, or average length is misreported.

Second, the abstract and introduction claim the method requires “about 1M label tokens,” while Appendix A.3.2 specifies “about 1.2M”. While close, this lack of precision, combined with the calculation error above, indicates a need for rigorous alignment between the summary claims and the detailed experimental logs. These issues are fixable by correcting the numbers to be mathematically consistent.

Action items.

- **[writing]** Section 5.1 claims 600 steps on 48K avg length, while Appendix A.3.2 claims 180M total tokens. With batch size 8, $600 \times 8 \times 48K = 230.4M$, contradicting 180M. Reconcile step count, batch size, or total token count to ensure internal consistency.
- **[writing]** Abstract claims ‘about 1M label tokens’ and ‘few hundred steps’, but Appendix A.3.2 specifies ‘1.2M label tokens’ and ‘600 steps’. Align the main text numbers with the appendix values or use consistent rounding to avoid minor numerical inconsistencies.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a compelling method for sparse attention, but the rhetoric in the Abstract, Introduction, and Conclusion frequently exceeds the scope of the demonstrated evidence.

1. Overgeneralization of “Solves” and “Near-Lossless” Claims The Abstract states the method “solves” the efficiency bottleneck and preserves “near-lossless accuracy.” The Introduction reinforces this by claiming the method “strictly preserves accuracy.” However, the experimental evidence (Tables 1-3) is derived exclusively from two specific model variants: Qwen3-Coder-30B-A3B and Qwen3-30B-A3B-Think. While the results on these models are strong, the language implies a universal solution for all long-context LLMs. The term “solves” is particularly strong for a method that still incurs some accuracy drop on certain sub-tasks (e.g., MMLU-Chem in Table 3 shows a drop from 88.00 to 87.80, and MMLU-CS drops from 86.10 to 85.12). The claim of “near-lossless” is only valid within the specific tested regime. *Fix:* Replace “solves” with “mitigates” or “addresses.” Qualify “near-lossless accuracy” to “near-lossless accuracy on the evaluated Qwen3 models and benchmarks.”

2. Unqualified “First-Ever” Claim The Introduction asserts: “To the best of our knowledge, RTPurbo is the first method to achieve such near-lossless compression with lightweight continual training.” This is a broad historical claim. The paper’s evidence only covers the Qwen3 family. Other methods (e.g., DuoAttention, RazorAttention) also utilize lightweight adaptation strategies on specific models. Without a comprehensive survey proving no other method has achieved similar results on *any* model, this “first” claim is overreaching. *Fix:* Narrow the claim to “the first method to achieve... on the Qwen3-Coder-30B-A3B model” or rephrase to “demonstrates that near-lossless compression is achievable with lightweight training,” removing the “first” assertion.

3. Conclusion Re-opens Scoped Limitations The Conclusion states: “full-attention models can already support effective sparse execution with lightweight post hoc adaptation.” This generalizes the finding to the entire class of “full-attention models.” The paper’s own Limitations section (Appendix) correctly identifies that the experiments “mainly focus on the Qwen3 family” and that broader validation is needed. However, the Conclusion ignores this boundary, presenting a specific finding as a general law of LLM architecture. This creates a disconnect between the careful limitations and the broad final takeaway. *Fix:* Modify the Conclusion to explicitly state

that this finding holds for the “Qwen3 family of models studied” or add a clause acknowledging that generalization to other architectures remains an open question.

4. Missing “In-the-Wild” Validation The Abstract and Introduction frame the results as applicable to “real-world” scenarios (e.g., “multi-turn dialogue, long-horizon reasoning”). The experiments are conducted on static benchmarks (LongBench, RULER, MMLU-PRO). While these are standard proxies, the language implies deployment readiness or validation in dynamic, real-world settings which were not tested. *Fix:* Ensure the text clarifies that the “real-world” applications are the *motivation* for the benchmarks, not the *setting* of the evaluation. Use phrases like “benchmarks simulating real-world tasks” rather than implying the method was tested in production.

The paper’s core contribution is solid, but the framing needs to be tightened to match the specific experimental boundaries (Qwen3 models, specific benchmarks) to avoid overclaiming universality.

Action items.

- **[writing]** Abstract/Intro claim the method ‘solves’ the bottleneck and is ‘near-lossless’ universally. Evidence is limited to two Qwen3 models on specific benchmarks. Replace ‘solves’ with ‘mitigates’ and scope ‘near-lossless’ to ‘on evaluated Qwen3 models and benchmarks’.
- **[writing]** Intro claims RTPurbo is the ‘first method’ to achieve near-lossless compression with lightweight training. Evidence only covers Qwen3 models; prior work exists on other models. Qualify to ‘first on Qwen3-Coder-30B-A3B’ or remove ‘first’ to avoid overclaiming.
- **[writing]** Conclusion states ‘full-attention models’ generally support sparse execution, but evidence is limited to Qwen3. Limitations section admits this scope. Narrow conclusion to ‘Qwen3 family studied’ or explicitly acknowledge untested generalization to other architectures.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a method for sparsifying attention mechanisms in large language models to improve inference efficiency. The work relies on standard public datasets (FineWeb, Dolma 3) and evaluates performance on established public benchmarks (LongBench, RULER, MMLU-PRO). The methodology involves analyzing intrinsic model properties (attention head specialization) and applying lightweight post-hoc training (self-distillation) to existing open-weight models (Qwen3 family).

From a safety and ethics perspective, the paper does not present foreseeable, non-trivial risks of harm that are unaddressed.

1. **Data & Privacy:** The training data sources are public web corpora with no indication of

personally identifiable information (PII) collection or human-subjects research requiring IRB approval. The evaluation benchmarks are standard, non-sensitive datasets.

2. **Dual-Use:** While the method improves inference efficiency (lowering cost/latency), this is a general capability improvement common to many optimization papers. It does not specifically lower the barrier to a concrete harmful capability (e.g., automated vulnerability discovery, biological synthesis, or targeted disinformation generation) in a way that requires unique mitigation beyond standard responsible AI practices. The paper does not release new models or datasets that could be directly weaponized.
3. **Bias & Fairness:** The paper reports accuracy on standard benchmarks but does not claim to solve fairness issues nor does it present results showing a new, unmitigated bias against a specific demographic group. The focus is on efficiency vs. accuracy trade-offs, which is within the scope of the reported experiments.

The paper includes a “Limitation” section (Appendix) acknowledging constraints on head specialization stability and evaluation scope. No specific safety disclosures, responsible release plans, or ethical statements are missing given the nature of the research (algorithmic optimization on public data). The work is low-risk by construction.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method for sparse attention, but the experimental design has specific gaps that prevent the evidence from fully supporting the claims of “near-lossless” accuracy and robust efficiency gains.

First, the accuracy results in Tables 1, 2, and 3 are presented as single-point estimates without any measure of variance (standard deviation, confidence intervals) or seed count. In long-context benchmarks, performance can fluctuate significantly based on the specific random seed or the composition of the test set. The reported improvements over strong baselines (e.g., the 0.94% gain over the top- k variant on LongBench) are small and fall well within the typical noise floor of LLM evaluation. Without reporting results across multiple seeds (e.g., 3-5 runs), it is impossible to distinguish a genuine methodological improvement from lucky initialization or test-set variance.

Second, the comparison between the proposed dynamic top- p method and the “static top- k ” baseline is potentially confounded by hyperparameter tuning asymmetry. The paper states $k = 4096$ was “empirically set,” but does not clarify if this value was tuned on a validation set or chosen post-hoc to maximize performance. If the top- k baseline was tuned while the top- p method used a fixed $p = 0.9$, the comparison is unfair. To isolate the benefit of the *dynamic* mechanism, the authors should either tune the top- k baseline on a held-out set or,

more rigorously, run an ablation where the top- p mechanism is forced to use a fixed budget equal to its average token count, ensuring the comparison is strictly about the selection logic, not the budget size.

Finally, the efficiency claims rely on micro-benchmarks of a single attention layer. While the 9.36x speedup for a single layer is impressive, it does not guarantee the same end-to-end speedup for the full model. Inference latency is often dominated by memory bandwidth, kernel launch overheads, or non-attention operations (e.g., MLP layers, normalization) which are not accelerated by the sparse attention kernel. The claim of “2.01x decode speedup” needs to be backed by end-to-end generation latency measurements on the full model, not just the attention operator, to rule out the possibility that the attention speedup is masked by other bottlenecks.

Action items.

- **[writing]** The paper presents a compelling method for sparse attention, but the experimental design has specific gaps that prevent the evidence from fully supporting the claims of “near-lossless” accuracy and robust efficiency gains. First, the accuracy results in Tables 1, 2, and 3 are presented as single-point estimates without any measure of variance (standard deviation, confidence intervals) or seed count. In long-context benchmarks, performance can fluctuate significantly based on the specific random

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is currently insufficient to support the precision of the claims made. While the experimental design (comparing against baselines) is sound, the **treatment of the resulting numbers** lacks necessary uncertainty quantification.

Specifically, Tables 1, 2, and 3 present accuracy metrics (e.g., “54.24%”, “90.06%”) as exact point estimates. In the context of LLM evaluation, where results can vary significantly based on random seeds, sampling temperature, or minor implementation details, reporting a single number without a standard deviation (SD), standard error (SE), or confidence interval (CI) is misleading. It implies a level of precision that is not statistically justified. The same issue applies to the speedup claims in the “Runtime Analysis” section; reporting “9.36×” without a range or variance across multiple timing runs suggests a deterministic outcome where one likely does not exist.

Furthermore, the paper highlights “best” results across numerous benchmarks (e.g., 16 sub-tasks in LongBench, 10 in RULER) without addressing the multiple comparisons problem. With so many pairwise comparisons, some “significant” improvements are expected by chance alone. While formal hypothesis testing is not always standard in every ML sub-field, the absence of *any* uncertainty reporting (even simple SD over seeds) prevents the reader from distinguishing between a robust improvement and a lucky run.

The fix is primarily a reporting one: re-run the experiments with multiple seeds (if not already done) and report mean $\pm$ SD, or explicitly state the single-run nature of the results and adjust the language to reflect that these are point estimates, not population parameters. No re-analysis of raw data is strictly required if the per-seed logs exist, but if only single runs were performed, the paper must acknowledge this limitation.

Action items.

- **[writing]** Tables 1-3 (LongBench, RULER, Reasoning) report single-point accuracy scores (e.g., ‘54.24%’) without any measure of uncertainty (SD, SE, or CI) or mention of the number of random seeds used. In deep learning, single-run results are not statistically robust. Report mean $\pm$ SD over at least 3 seeds for all reported metrics, or explicitly state that results are from a single run and treat them as such in the text.
- **[writing]** Section 5.1 claims ‘up to a $9.36\times$ prefill speedup’ and ‘ $2.01\times$ decode speedup’ based on single measurements. Speedup ratios in systems papers are highly sensitive to hardware noise and warm-up effects. Report these speedups as mean $\pm$ SD over multiple runs (e.g., 5-10 iterations) to establish statistical stability, rather than presenting a single point estimate as a definitive property.
- **[writing]** Table 2 (RULER) and Table 3 (Reasoning) compare RTPurbo against 5+ baselines across multiple sub-tasks (e.g., 10+ columns in RULER). The paper highlights ‘best’ results with bolding but performs no statistical significance testing (e.g., paired t-tests or bootstrap) to determine if the observed differences are real or due to random variance. At minimum, add a note acknowledging that without multiplicity correction or significance testing, the ‘best’ labels may reflect noise.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the prose is clear, but there are specific instances where sentence construction and structural flow impede immediate comprehension.

In Section 2.2, the subsection begins with a sentence fragment (“Retrieval heads should assign high attention...”) that fails to function as a complete thought, forcing the reader to wait for the next sentence to find the main verb. This breaks the momentum of the argument. Similarly, in Section 3.2, the explanation of sparsity in MQA/GQA models uses the phrase “our head partition” without clearly defining what “partition” refers to in that specific context, creating a momentary ambiguity about whether it refers to the model architecture or the specific selection of heads.

A more significant flow issue appears in Section 4, where a sentence describing input characteristics contains a broken mathematical expression (“ $\$0.93$ ”). This looks like a LaTeX compilation

error or a missing variable that disrupts the reading of the results. Finally, the abstract, while informative, ends with a generic concluding sentence that misses an opportunity to reiterate the specific method (head-wise sparsification) and the primary quantitative result (9.36x speedup) that the body of the paper emphasizes. Tightening the abstract to include these specific details would ensure it serves as a robust standalone summary.

These issues are minor and fixable through careful editing, but they currently create small friction points for the reader.

Action items.

- **[writing]** The paper is generally well-structured and the prose is clear, but there are specific instances where sentence construction and structural flow impede immediate comprehension. In Section 2.2, the subsection begins with a sentence fragment (“Retrieval heads should assign high attention...”) that fails to function as a complete thought, forcing the reader to wait for the next sentence to find the main verb. This breaks the momentum of the argument. Similarly, in Section 3.2, the explanation of spa